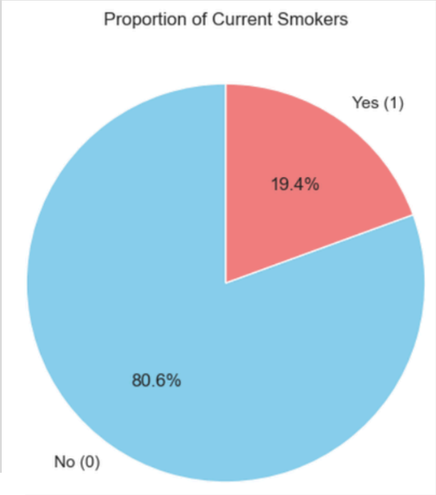
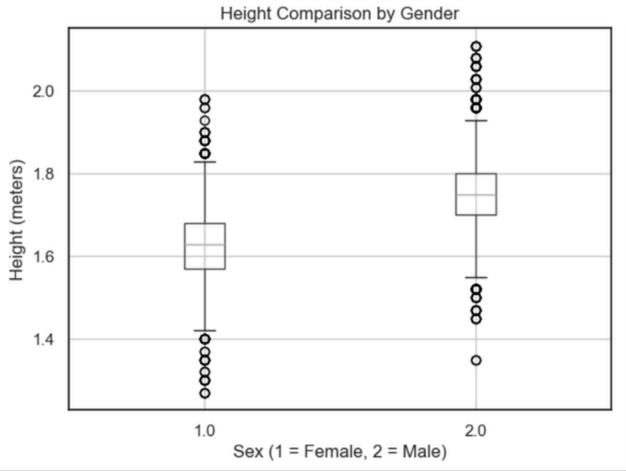
HOW DO GENDER AND HEIGHT JOINTLY INFLUENCE THE PROPORTION OF STUDENTS WHO SMOKE?

性別與身高如何交互影響學生的抽菸比例？

1.Overall Smoking Proportion

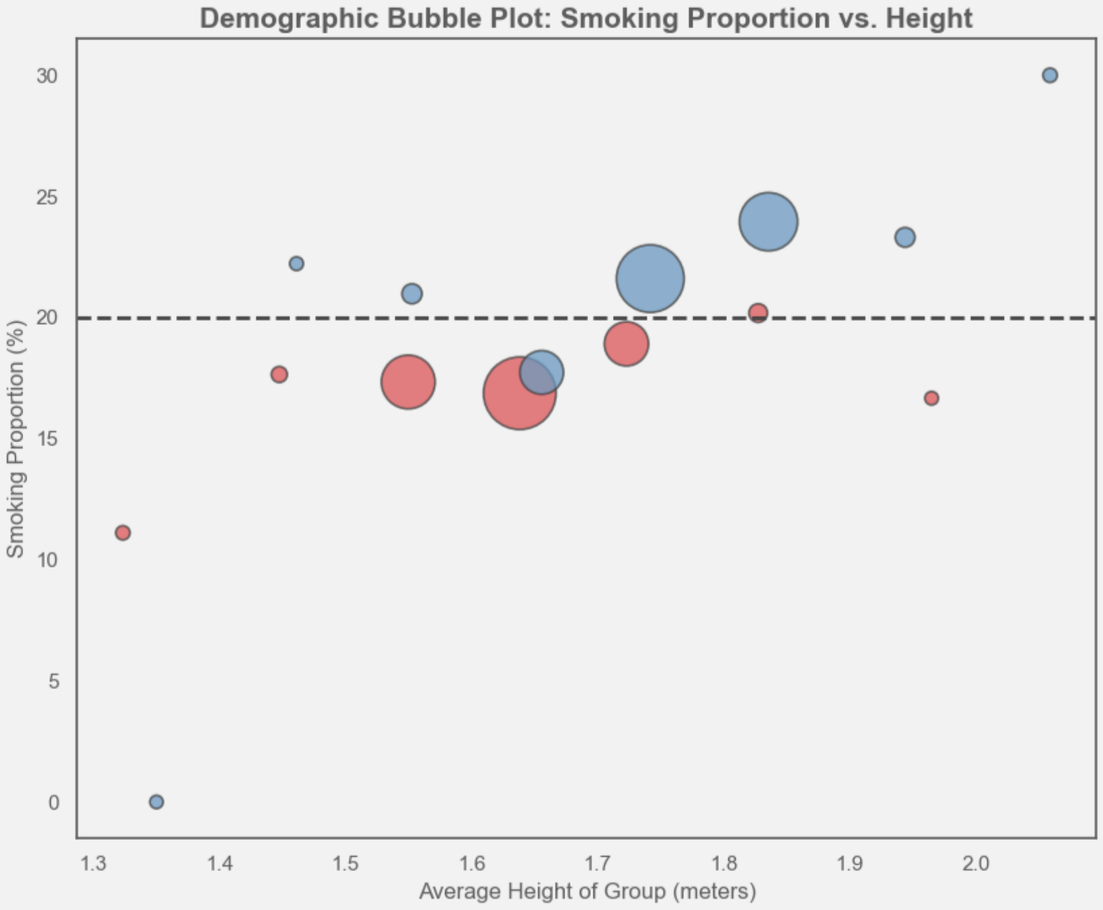
The actual smoking rate is 19.4%, providing a baseline for subgroup comparisons.

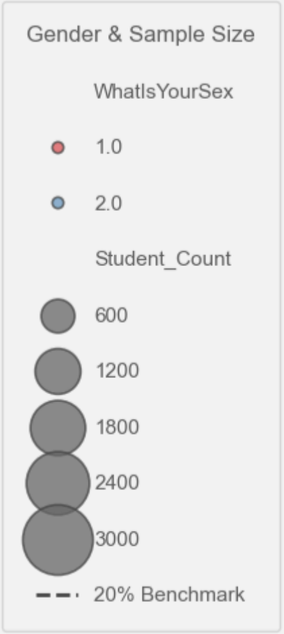




2.Height Distribution by Gender

Significant height differences between genders justify the need for segmented behavior analysis.





3.we tested smoking rates and height against population benchmarks.

	Analysis	Sample Size	Estimate	p-value	Decision (alpha=0.05)
0	Proportion	13323	0.194326	1.015422e-01	Fail to reject H0
1	Mean	13062	1.694038	1.947176e-11	Reject H0

Advanced Interaction Analysis

Key Discovery: Taller males show a "step-wise" increase in smoking rates, while females remain stable.

Final Insight:

By filtering out statistical noise via bubble scaling, we identified "taller males" as a specific high-risk group that requires targeted health intervention.